

Project Proposal

Stargaze

Summary

I am working on an interactive project about the humans look at the night sky; every celestial objects astronomical parameters (light spectrum, mass, altitude... etc.) are translated into musical parameters as the observer sight wanders around a projection of the firmament. In real time, trajectories are displayed, and melodies arise from the mere act of observing the stars. From here on, an originally passive perception may move towards a conscious interaction with the system. Long story short, the project is a reflexion on possibly the oldest act of conscious observation deprived from any practical or survival benefit. It is arguably, the first purely aesthetic experience early humans indulged in.

Resources

I currently use accelerometers and motion tracking to allow the observer to move around the projection. It would be of great interest to use eye-tracking instead.

Flow of information

The project relies on many different layers, only one of which is related to music. There are other aspects involved such as perception, neural networks and visualisation, to name a few.

1. Measurements (fixation, saccade, coordinates)
2. Input processing → 1st audio response
3. Neural network (multi-layered perceptron, using backpropagation)
4. Output processing → 2nd audio response
5. Graphic interface
6. Collect data.

“Stargaze narrative” (subjective starspotting)

- A. Particular locations (stars) can be linked to particular interval/pitch information according to the ARICNS table.
- B. Alternatively, secondary information (jump length, orientation...) could be translated to musical parameters. Sonification of plastic arts: similarly to melodies in a musical texture, the eye-trajectory outlines the ‘melos’ of a painting. Color values surrounding the fixation cluster may convey the ‘harmonic field’ through which the ‘melos’ proceeds.

LENGTH of the saccade (straight-line distance from A to B)	
Proportion $LENGTH_0 / LENGTH_{-1}$	
Accumulated $LENGTH_{\sum 0 \dots x}$	
Curvature of the saccade trajectory ($180^\circ < > -180^\circ$)	
Derivation between $CURVE_0 / CURVE_{-1}$	
Accumulated $CURVE_{\sum 0 \dots x}$	
Orientation of the saccade (360 degrees)	
Inflexion between $ANGLE_0 / ANGLE_{-1}$	
Eccentricity (distance to the 0.0 coordinates)	
Time of rest (fixation)	
Time of the saccade movement	
Speed of the saccade (LENGTH / TIME)	
Overall pace (mean value of the time between movements) $\Sigma\{TIME(saccade), TIME(fixation)\}$	